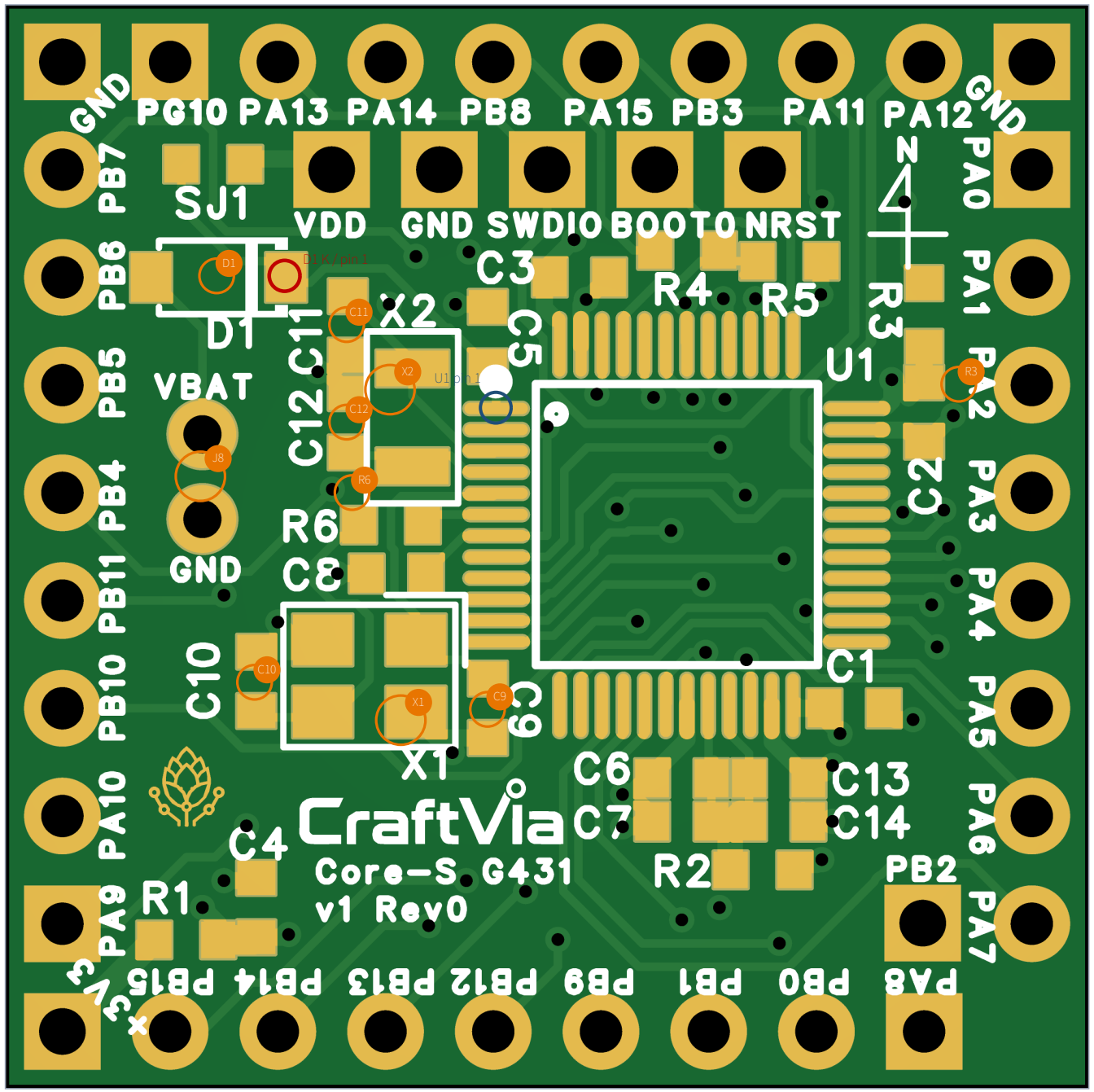


Assembly status

Variant code: P10 + V0 + C0

All mounted components are on the Top side.

DNP - 10 parts

C9, C10, C11, C12, D1, J8, R3, R6, X1, X2

Orange rings on the Top view identify every DNP location.

VBAT option / jumper

標準構成: SJ1 fitted。D1 / J8 はDNP。

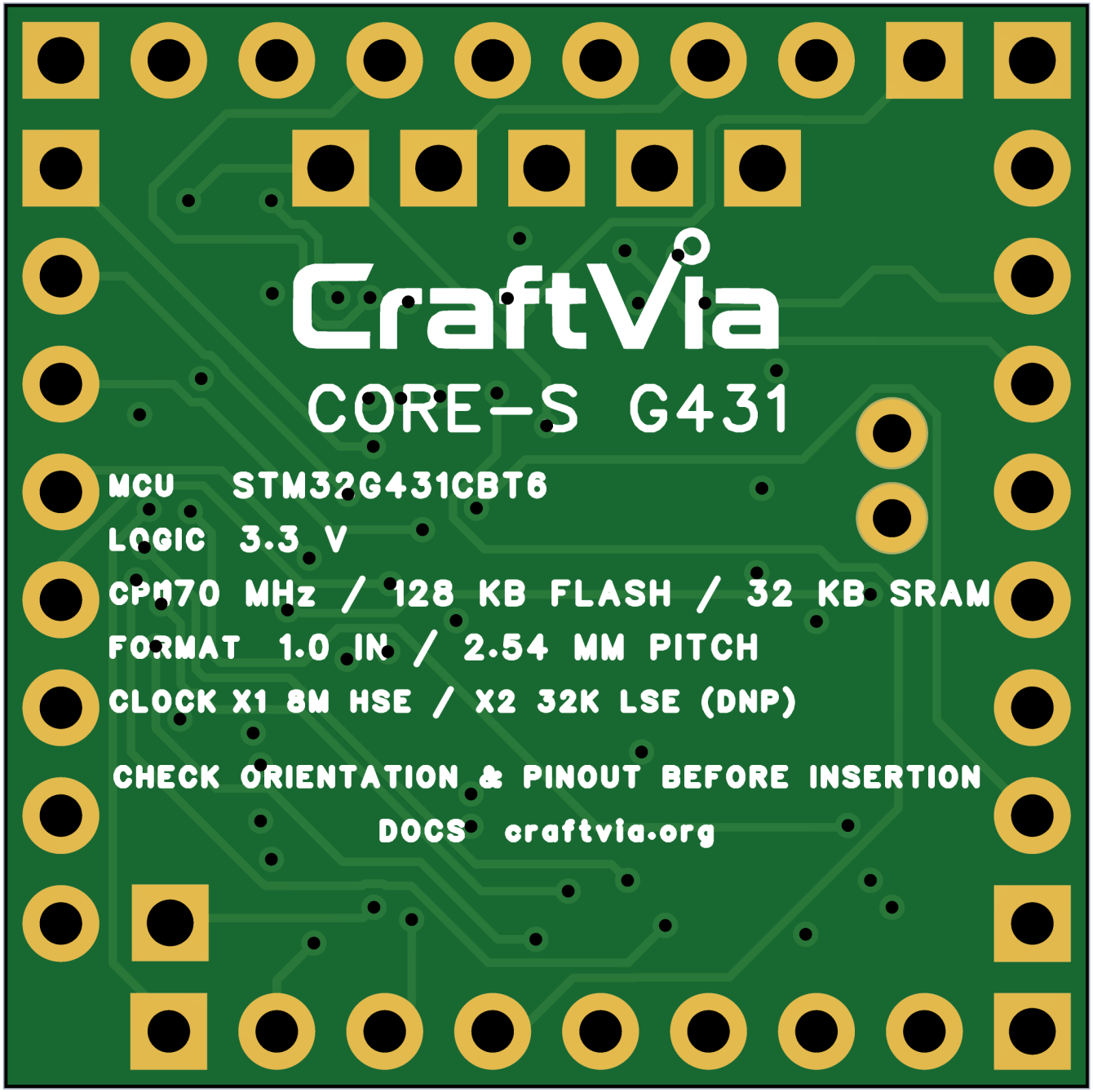
外部VBAT使用時: SJ1を外し、D1とJ8を実装する。

D1: pin 1 / K = VBAT_BACKUP側（Top viewで右・基板中央側）。pin 2 / A = J8 VBAT+側（左・基板端側）。

Orientation checks

U1の1番ピンはTop view左上。青丸と基板シルクの丸印を一致させる。

D1実装時は部品のカソード帯をK / pin 1側へ合わせる。



View convention

下面から基板を見た図

Viewed from the board underside.

上面からの透視図ではない。反転せず、この向きで照合する。

Bottom-side assembly

Rev0のBottom側に実装部品はない。

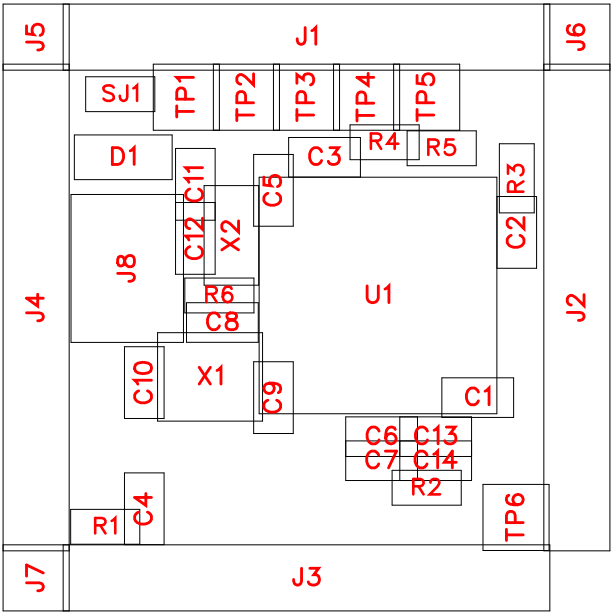
スルーホール部品のリード、はんだブリッジ、未はんだ、異物を検査する。

Cross-check

Top viewのDNP 10点とCPL除外対象が一致していることを確認する。

TP1-TP6は監査用CPLから除外。

製品: CraftVia Core-S G431 v1 Rev0
作成日: 2026-10-26
Variant code: P10 + V0 + C0

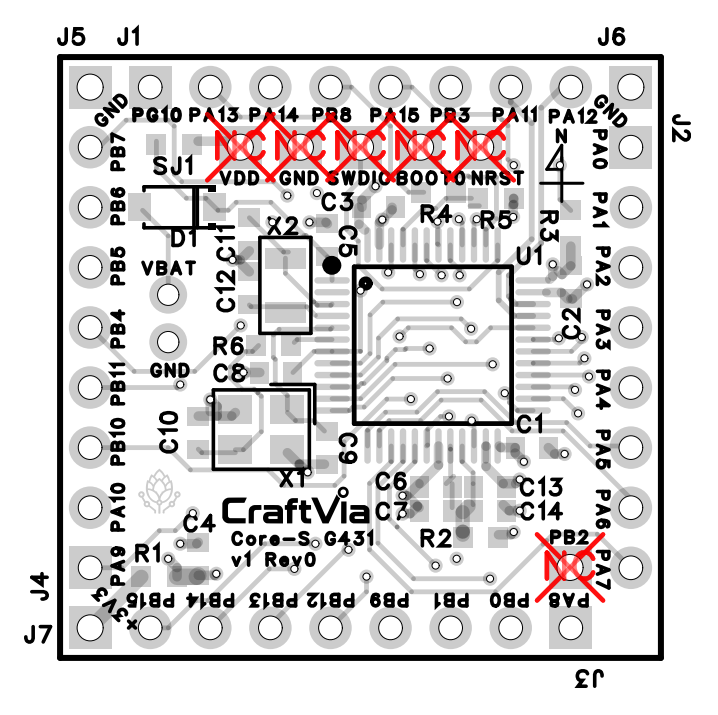

J5
J1
J6
SJ1
TP1
TP2
TP3
TP4
TP5
D1
C3
R4
R5
C11
C5
R3
C12
X2
C2
J8
U1
J4
J2
R6
C8
C10
X1
C9
C1
C6
C13
C7
C14
R2
C4
R1
TP6
J7
J3

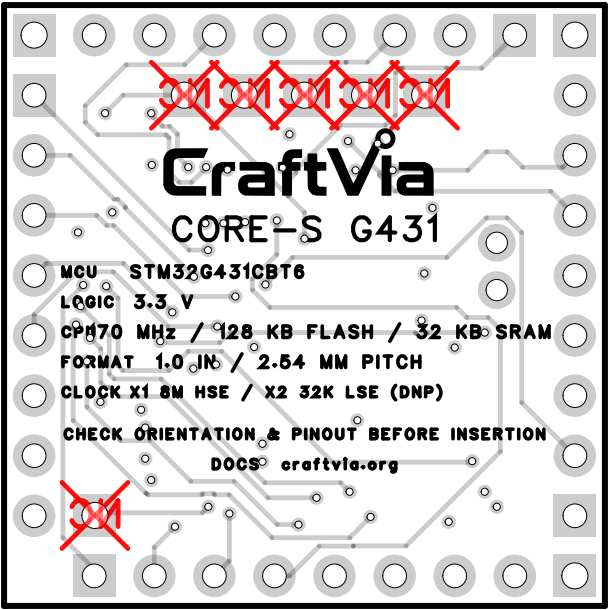
No.	Quantity	Comment	Designator	Footprint	Value	Manufacturer Part	Manufacturer	Supplier Part	Supplier	LCSC Part Name
1	3	100nF VDD	C1,C2,C3	C0603	100nF	CC0603KRX7R9BB104	YAGEO(国巨)	CC0603KRX7R9BB104.1	LCSC	
2	1	4.7uF bulk	C4	C0603	4.7uF	CL10A475KO8NNNC	SAMSUNG(三星)	CL10A475KO8NNNC.1	LCSC	
3	1	100nF VBAT	C5	C0603	100nF	CC0603KRX7R9BB104	YAGEO(国巨)	CC0603KRX7R9BB104.1	LCSC	
4	1	100nF VREF+	C6	C0603	100nF	CC0603KRX7R9BB104	YAGEO(国巨)	CC0603KRX7R9BB104.1	LCSC	
5	1	1uF VREF+	C7	C0603	1uF	CL10A105KB8NNNC	SAMSUNG(三星)	CL10A105KB8NNNC.1	LCSC	
6	1	100nF NRST	C8	C0603	100nF	CC0603KRX7R9BB104	YAGEO(国巨)	CC0603KRX7R9BB104.1	LCSC	
7	2	18pF HSE load (DNP, tune)	C9,C10	C0603	18pF	CL10C180JB8NNNC	SAMSUNG(三星)	CL10C180JB8NNNC.1	LCSC	
8	2	10pF LSE load (DNP, tune)	C11,C12	C0603	10pF	CL10C100JB8NNNC	SAMSUNG(三星)	CL10C100JB8NNNC.1	LCSC	
9	1	100nF VDDA	C13	C0603	100nF	CC0603KRX7R9BB104	YAGEO(国巨)	C14663	LCSC	100nF ±10% 50V
10	1	1uF VDDA	C14	C0603	1uF	CL10A105KB8NNNC	SAMSUNG(三星)	CL10A105KB8NNNC.1	LCSC	
11	1	BAS116GWJ VBAT isolation (DNP)	D1	SOD-123_L2.8-W1.6-LS3.7-RD		BAS116GWJ	Nexperia(安世)	BAS116GWJ.1	LCSC	85V 215mA
12	1	N 1-8	J1	HDR-TH_8P-P2.54-V-M		直排针2.54mm 1*8P		Header2.54mm 1*8P.1	LCSC	
13	1	E 1-8	J2	HDR-TH_8P-P2.54-V-M		直排针2.54mm 1*8P		Header2.54mm 1*8P.1	LCSC	
14	1	S 1-8	J3	HDR-TH_8P-P2.54-V-M		直排针2.54mm 1*8P		Header2.54mm 1*8P.1	LCSC	
15	1	W 1-8	J4	HDR-TH_8P-P2.54-V-M		直排针2.54mm 1*8P		Header2.54mm 1*8P.1	LCSC	
16	1	K-NW	J5	HDR-TH_1P-P2.54-V-M		2.54-1*1P针	BOOMELE(博穆精密)	Header-Male-2.54_1x1.1	LCSC	
17	1	K-NE	J6	HDR-TH_1P-P2.54-V-M		2.54-1*1P针	BOOMELE(博穆精密)	Header-Male-2.54_1x1.1	LCSC	
18	1	K-SW	J7	HDR-TH_1P-P2.54-V-M		2.54-1*1P针	BOOMELE(博穆精密)	Header-Male-2.54_1x1.1	LCSC	
19	1	VBAT IN (PH 2.0mm)	J8	CONN-TH_B2B-PH-K-S		XY-B2B-PH-K-S	XYECONN(辛译)	XY-B2B-PH-K-S.1	LCSC	
20	1	0R / FB VDD	R1	R0603	0R	0603WAF0000T5E	UNI-ROYAL(厚声)	0603WAF0000T5E.1	LCSC	
21	1	0R / FB VDDA	R2	R0603	0R	0603WAF0000T5E	UNI-ROYAL(厚声)	0603WAF0000T5E.1	LCSC	

22	1	10k NRST pull-up (DNP)	R3	R0603	10k	0603WAF1002T5E	UNI-ROYAL(厚声)	0603WAF1002T5E.1	LCSC	
23	1	1k BOOT series	R4	R0603	1k	0603WAF1001T5E	UNI-ROYAL(厚声)	0603WAF1001T5E.1	LCSC	
24	1	10k BOOT pull-down	R5	R0603	10k	0603WAF1002T5E	UNI-ROYAL(厚声)	0603WAF1002T5E.1	LCSC	
25	1	0R HSE REXT start (DNP, tune)	R6	R0603	0R	0603WAF0000T5E	UNI-ROYAL(厚声)	0603WAF0000T5E.1	LCSC	
26	1	0R VBAT link (default fitted)	SJ1	R0603	0R	0603WAF0000T5E	UNI-ROYAL(厚声)	0603WAF0000T5E.1	LCSC	
27	1	STM32G431CBT6	U1	LQFP-48_L7.0-W7.0-P0.50-LS9.0-BL		STM32G431CBT6	ST(意法半导体)	STM32G431CBT6.1	LCSC	
28	1	8MHz HSE: SX32Y008000BC1T001 (DNP/Optional)	X1	CRYSTAL-SMD_4P-L3.2-W2.5-BL	8.000MHz	SX32Y008000BC1T001	TKD(泰晶)	SX32Y008000BC1T001.1	LCSC	
29	1	32.768kHz LSE: X1A0001410001 (DNP/Optional)	X2	OSC-SMD_L3.2-W1.5	32.768kHz	X1A0001410001	EPSON(爱普生)	FC-135R32.768kHz7.0pF20PPM.1	LCSC	

Description	Mounting Type
VREF+ local decoupling; place at U1 pin 20 with shortest return to VSSA pin 19	
VREF+ local bulk decoupling; place at U1 pin 20 with shortest return to VSSA pin 19	
HSE load capacitor nominal start value for X1 CL=12pF; tune for PCB stray capacitance and startup margin	
LSE load capacitor nominal start value for X2 CL=7pF; tune for PCB stray capacitance and startup margin	
VDDA local decoupling; place at U1 pin 21 with shortest return to VSSA pin 19	
VDDA local bulk decoupling; place at U1 pin 21 with shortest return to VSSA pin 19	
Optional low-leakage series isolation diode for external VBAT; populate for direct primary-cell use; anode to J8 VBAT+, cathode to VBAT_BACKUP	
External VBAT input. Populate D1 and remove SJ1 before connecting a primary cell.	
Optional external NRST pull-up; standard assembly DNP	
Series REXT on OSC_OUT; start with 0R for bring-up, then verify drive level and negative-resistance margin	
Default fitted: VBAT follows VDD_CORE; remove SJ1 and populate D1 before applying external VBAT via J8	

Primary HSE crystal, SMD 3225 4-pin; standard assembly DNP; validate load, startup and drive level on PCB	
Primary LSE crystal, Epson FC-135R 3215; standard assembly DNP; tune load capacitors and configure LSEDRV	


J5
J1
J6
J2
J4
J7
J3
GND
PG10
PA13
PA14
PB8
PA15
PB3
PA11
PA12
GND
PB7
PA0
PB6
PA1
PB5
PA2
PB4
PA3
PB11
PA4
PB10
PA5
PA10
PA6
PA9
PA7
3V3
PB16
PB14
PB13
PB12
PB9
PB1
PB0
PA8
PB2
SJ1
VDD
GND
SWDIO
BOOT0
NRST
D1
VBAT
C3
X2
C11
C12
C5
R4
R5
R3
U1
C2
R6
GND
C8
C10
C9
C1
X1
C6
C13
C7
C14
C4
CraftVia
Core-S_G431
v1 Rev0
R1
R2


CraftVia
CORE-S G431
MCU STM32G431CBT6
LOGIC 3.3 V
CPU 170 MHz / 128 KB FLASH / 32 KB SRAM
FORMAT 1.0 IN / 2.54 MM PITCH
CLOCK X1 8M HSE / X2 32K LSE (DNP)
CHECK ORIENTATION & PINOUT BEFORE INSERTION
DOCS craftvia.org

